Mahtab *Baayen*

AI / ML RESEARCHER & DATA SCIENTIST

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Machine Learning Computer Vision NLP & LLMs

Data Science & Engineering Traffic & Transportation Control Systems

Researcher and AI/ML specialist with a M.Sc. in Artificial Intelligence (Maastricht University) and extensive Ph.D.-level research experience at TU Delft. Author of 7 peer-reviewed publications in IEEE ITSC, TRB, and TGF. Patent holder. Over 10 years bridging fundamental research and real-world applications in machine learning, computer vision, NLP, and traffic flow modelling.

RESEARCH & PROFESSIONAL EXPERIENCE

System Engineer
June 2025 — Present

Maestronic

- Design and implement embedded systems for industrial applications
- Develop real-time computer vision solutions for manufacturing quality control
- Integrate AI/ML models into edge computing devices for automated inspection

Data Scientist / AI Consultant
Sep 2024 — Jun 2025

Conclusion Intelligence — Consultant for APG Pensionfonds, Heerlen

- Developed knowledge graphs and ontologies for Dutch pension regulations using NLP and LLMs
- Built automated information extraction pipelines processing complex legal documents

EDUCATION

M.Sc. Artificial Intelligence
Maastricht University

2021 — 2024

Thesis: Online shockwave detection and tracking in vehicular traffic using deep learning

Advisor: Dr. M. Popa

M.Sc. Computational Engineering
Ruhr University Bochum, Germany

2006 — 2010

Thesis: Identification of Material Properties using Extended Kalman Filter within Finite Element Method

Advisor: Prof. Dr.-Ing. T. Nestorovic

B.Sc. Water Engineering
University of Tehran, Iran

- Implemented semantic search and query systems for regulatory compliance

M.Sc. Research — Artificial Intelligence
2021 — 2024

Maastricht University — Department of Data Science

- Deep learning models for real-time shockwave detection and tracking in traffic video streams
- Computer vision pipeline extracting traffic flow heterogeneity metrics from surveillance footage
- Published research on CNN and LSTM architectures for traffic pattern recognition

Thesis: Online shockwave detection and tracking in vehicular traffic using deep learning

Internship: Deriving traffic flow heterogeneity from video data

Advisor: Dr. M. Popa

Traffic Systems Consultant
2018 — 2019

Sweco

- Developed and tested adaptive traffic control software improving flow by 15–20%
- Deployed intelligent transportation systems (ITS) for major infrastructure projects
- Authored commercial and scientific publications

R&D Researcher
2016 — 2018

Dynniq

- R&D of intelligent transportation solutions (ITS)
- Invented and patented a green wave traffic control system (Patent granted 2018)
- Coordinated multiple EU-funded research projects and grant applications

Ph.D. Candidate
2011 — 2016

TU Delft — Department of Traffic & Transportation

- Developed a unified multi-scale traffic flow modelling framework capable of reproducing key traffic phenomena with high accuracy
- Published 7 peer-reviewed papers across IEEE ITSC, TRB, TGF, and IFORS
- Best Paper nomination at 16th IEEE ITSC (2013); Best Poster nomination at TRAIL (2012)

2002 — 2006

Thesis: Underground Dams

Advisor: Prof. Habibi

TECHNICAL SKILLS

PROGRAMMING

Python | R | C++ | Java | Go | Bash | SQL

DATA SCIENCE & AI

TensorFlow | PyTorch | Keras | scikit-learn | OpenCV | NLTK | Hugging Face | Pandas | NumPy

BACK-END & APIS

FastAPI | Flask | Django | Node.js

CLOUD & DEVOPS

AWS | Azure | Docker | Kubernetes | Jenkins | CI/CD

DATA ENGINEERING

Apache Spark | Hadoop | Airflow | Kafka | ETL Pipelines

DATABASES

PostgreSQL | MySQL | SQLite

TOOLS

— Received NWO Ph.D. Grant (2011)

Dissertation: Multi-Scale Traffic Flow Modelling and Simulation

Advisors: Prof. Serge Hoogendoorn, Prof. Hans van Lint

Research Intern
2008 — 2009

EUTech, Aachen, Germany

— Optimized numerical algorithms for solving non-linear thermodynamic state equations

PUBLICATIONS & PATENT

A system for controlling traffic lights and a corresponding control unit
Mahtab Joueiai, Robbin Blokpoel
Patent granted — October 2018

Bicycle modelling in SUMO for accurate traffic simulation
Robbin Blokpoel, Mahtab Joueiai
SUMO Conference — May 2016

Discrete Phenomena-Based Multi-Scale Traffic Flow Modelling
Mahtab Joueiai, Hans van Lint, Serge Hoogendoorn
Traffic and Granular Flow (TGF) — October 2015

A Multi-Scale Traffic Flow Model Based on the Mesoscopic LWR Model
Mahtab Joueiai, Ludovic Leclercq, Hans van Lint, Serge Hoogendoorn
TRB 94th Annual Meeting — January 2015

Scale Selection in Multi-Scale Traffic Flow Modeling
Mahtab Joueiai
20th IFORS Conference — July 2014

Emergency Evacuation of Tehran City: Simulation results
Mahtab Joueiai, Hans van Lint, Serge Hoogendoorn, Nima Pouya
13th International Conference, Tehran — February 2014

Multi-Scale Traffic Flow Modeling In Mixed Networks
Mahtab Joueiai, Hans van Lint, Serge Hoogendoorn
TRB 93rd Annual Meeting — January 2014

Git / GitHub | Jupyter | LaTeX | Linux | Tableau | Power BI

LANGUAGES

Persian	Native
English	Advanced
Dutch	C2
German	Intermediate
Arabic	Basic

Generic solutions for consistency problems in multi-scale traffic flow models — analysis and preliminary results

Mahtab Joueiai, Hans van Lint, Serge Hoogendoorn

16th IEEE Conference on Intelligent Transportation Systems (ITSC) — October 2013

TEACHING & ACADEMIC SERVICE

PhD Meeting Organiser — 2013 — 2014

TU Delft

- Organised biweekly PhD meetings for the Traffic & Transportation research group

Teaching Assistant — Traffic Flow Theory — 2011 — 2012

TU Delft — Master's course

- Assisted in teaching the master-level course on Traffic Flow Theory

AWARDS & GRANTS

Best Paper Nomination — 16th International IEEE Conference on Intelligent Transportation Systems (2013)

Best Poster Nomination — TRAIL Conference (2012)

NWO Ph.D. Grant — Netherlands Organisation for Scientific Research (2011)